
Herd Immunity and Learning Externalities in Markets of Adaptive Models

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Abstract

1 A learning agent can satisfy every single-agent stability criterion and still help
2 destabilize the market it operates in. When adaptive agents share an environment
3 that reacts to their decisions, each agent’s retraining reshapes the data every other
4 agent learns from next, a *learning externality* that no agent internalizes. Its strength
5 is governed not by the number of firms but by the *effective number of independent*
6 *learners*, the leading eigenvalue of the correlation matrix of firms’ response Jaco-
7 bians, so foundation-model concentration enters the stability condition as a term
8 rather than as a concern: fifty firms fine-tuning one vendor’s model are dynami-
9 cally close to one large learner. Three closed-form levers follow from a reduction
10 lemma we derive in the main text. A crowding-cadence frontier bounds how often
11 a market can retrain and closes past a critical crowding level. A herd-immunity
12 threshold makes the market stable exactly when the corrected fraction exceeds
13 $(1 - 1/m_N)/e$ at correction efficacy e , the imperfect-vaccine coverage law with
14 the systemic modulus as the reproduction number. A Pigouvian wedge prices the
15 externality, and every market of two or more firms over-adapts. The last two levers
16 are substitutes along a frontier we compute. In a shared-pool order-flow market
17 the mechanism reproduces a published $1.74\times$ and $3.16\times$ amplification at two and
18 three firms, bit for bit.

19 1 Introduction

20 A learning agent can pass every stability test we know how to run and still help destabilize the market
21 it operates in.

22 The test we know how to run is a single-agent test. A model is evaluated, stress-tested and certified
23 alone, against a distribution treated as given. When the deployed model reshapes that distribution, the
24 performative prediction literature supplies the correction: repeated retraining converges if and only if
25 a modulus $m = \varepsilon\beta/\gamma$ stays below one, where ε measures how strongly the environment responds
26 and γ is the learner’s own objective curvature (Perdomo et al., 2020). Certify $m < 1$ and the loop is
27 safe.

28 It is not, once the agent has company. Consider dealers whose quotes reshape a common pool
29 of informed flow, recommenders competing for one attention pool, or lenders pricing against one
30 population of borrowers. Each agent’s retraining changes the data every *other* agent will learn from
31 next, and no agent internalizes that effect. We call this a **learning externality**: every participant
32 behaves rationally, every participant passes its own test, and the interaction amplifies a loop none
33 of them chose. The intuition is not new. Banks that each diversify optimally hold similar portfolios
34 and fail together (Beale et al., 2011; Wagner, 2010), and funds running similar strategies unwound
35 simultaneously in August 2007 with the crowding invisible from any single risk report (Khandani

and Lo, 2007). What is new is that the agents now *learn*, which changes both what determines the danger and what can be done about it.

What determines it is not how many firms there are. Let firm i have a response Jacobian E_i describing how its own deployment reshapes the flow it faces, and let R be the correlation matrix of the vectorized E_i . We show the market’s stability is governed by $m_N = N_{\text{eff}} m_1$ with $N_{\text{eff}} = 1 + \kappa(\lambda_{\max}(R) - 1)$, where κ is the spillover between firms and $\lambda_{\max}(R)$ is the **effective number of independent learners**: a hundred firms perturbing the environment in a hundred orthogonal directions are dynamically one firm. With s the fraction of each response attributable to a common model, vendor or pretraining corpus, this reads $m_N = m_1(1 + \kappa s(N - 1))$, so **the effective number of learners is the number of independent models, not the number of firms**. The quantity a competition regulator measures is the wrong one: fifty equal-share firms have a minimal Herfindahl index and look perfectly competitive, and if all fifty share one vendor they are dynamically a monoculture (Hirschman, 1964).

What can be done about it takes three forms, and two are substitutes. Because every result is stated in m_N , each inherits the supply chain for free. *Slow down*: K -step retraining faces a window $K < K_{\max}(m_N)$ that narrows as crowding rises and closes past a critical crowding level beyond which no frequency is safe. *Correct*: feedback-aware updates (Izzo et al., 2021) run inside the market produce exactly the epidemiological herd-immunity law with the systemic modulus as the reproduction number. *Diversify*: that threshold rises in s , so a market reaches stability along either axis, and the iso-stability frontier between the two is this paper’s most directly usable object. Pricing the externality completes the policy triple, and the decentralized equilibrium over-adapts for every $N \geq 2$.

Contributions. (1) The effective number of independent learners as the reproduction number of a market of adaptive models, derived from a reduction lemma, turning foundation-model concentration into a term in a stability condition, with two counterexamples invalidating mean-similarity diversity indices. (2) The crowding-cadence frontier and its critical crowding level, composing lazy deployment with multi-agent amplification. (3) A mixed-market herd-immunity theorem, exact at finite correction strength, whose clean form is the imperfect-vaccine coverage law and which carries a critical efficacy. (4) The substitution frontier between diversity and corrected learning, which exists only once (1) and (3) are expressed in one quantity. (5) The Pigouvian wedge and the over-adaptation corollary.

2 Related work

Monoculture, and why it is static. Kleinberg and Raghavan (2021) show that when many firms adopt the same algorithm welfare can fall even when the shared algorithm is the better one, because correlated selection destroys the option value of independent evaluation; Bommasani et al. (2022) formalize systemic exclusion when many deployments inherit one model. Both are cross-sectional, measured at one point in time on one cohort, so neither can express a stability boundary or a rate. A market can be statically fine and dynamically fragile at the same shared-model fraction.

Multiplayer performative prediction, and the containment relation. Narang et al. (2023) is the nearest dynamics: multiple learners in a shared decision-dependent environment, with equilibrium concepts and convergence conditions for repeated retraining. The containment relation is worth stating, since Theorem 2 should not have to be taken on trust as more than a corollary. **Theirs bounds scalars**: the sensitivity of the joint distribution map to the players’ decisions, measured by Lipschitz constants, against their objective curvature, uniformly over the joint action. **Ours bounds a spectral radius**, $m_N = m_1[1 + \kappa(\lambda_{\max}(R) - 1)]$, in which those scalars appear only inside m_1 . **Ours is strictly sharper on the class their constants describe, and is not implied by theirs**. Sharper, because at fixed constants N_{eff} still ranges over $[1, 1 + \kappa(N - 1)]$, so a uniform bound must hold at the worst alignment structure while ours need not, and every market between the two is one their condition rejects and ours admits. Not implied, because nothing in their framework represents *which direction* each agent perturbs the environment: two markets with identical constants and opposite alignment structure are indistinguishable to them and differ in stability for us. The substitution $\mathbf{1}\mathbf{1}^\top \rightarrow R$ adds an object rather than tightening a constant, and that object, not the dynamics, carries monoculture, provenance, the supply chain and every policy statement below. The single-learner

88 lazy-deployment slope of Mendler-Dünner et al. (2020) is what Theorem 3 composes with multi-agent
 89 amplification, a composition not previously made; the corrected update of Izzo et al. (2021) is what
 90 Theorem 4 puts into the game, having only ever been run single-learner.

91 **Systemic risk, epidemiology, and the statistic.** Beale et al. (2011) asked this question of banks,
 92 showing individually prudent portfolio choices can be collectively destabilizing; their agents do not
 93 learn, so the homogeneity is a portfolio overlap rather than an alignment of feedback directions,
 94 and there is no second instrument to trade diversity against. Diekmann et al. (1990) converts the
 95 epidemiological parallel from analogy into identity: R_0 in a heterogeneous population is *defined* as
 96 the spectral radius of a next-generation operator, and m_N is the spectral radius of a linearized joint
 97 retraining operator, so the two thresholds are the same statement and the vaccination law transfers
 98 with its refinements. The largest eigenvalue of a correlation matrix as an effective count is meanwhile
 99 standard in random-matrix finance (Laloux et al., 1999; Plerou et al., 2002), signal processing (Roy
 100 and Vetterli, 2007) and ecology (Hill, 1973); the novelty is the matrix it is computed from, response
 101 Jacobians rather than returns, and the condition it enters. Work reading macroeconomic state off
 102 public sentiment (Nagarajan and Mittal, 2024) estimates through an exogenous channel, whose
 103 accuracy does not depend on the state of the system being measured, while the estimator of Section 9
 104 reads an endogenous one that sharpens as the market approaches instability, so the two face opposite
 105 sample-complexity problems near the event of interest.

106 3 Setup: private against systemic stability

107 **The base result.** Consider N symmetric dealers quoting the same instruments and sharing one
 108 pool of informed order flow, coupled by a spillover $\kappa \in [0, 1]$: one dealer tightening its quotes
 109 raises the toxic flow routed to every dealer, in proportion κ . Each periodically retrains on the flow
 110 it has observed and redeploys, and a firm in isolation is stable exactly when its own performative
 111 modulus satisfies $m_1 < 1$. The joint retraining map has Jacobian $J = -m_1[(1 - \kappa)I + \kappa \mathbf{1}\mathbf{1}^\top]$,
 112 whose dominant common mode sits at $-m_1 N_{\text{eff}}$ with $N_{\text{eff}} = 1 + \kappa(N - 1)$, so the market crosses
 113 into instability a factor N_{eff} earlier than any individual dealer’s own loop would. Measured in a
 114 shared-pool simulator with real spreads, inventory and liquidity feedback, amplification is $1.74\times$ at
 115 $N = 2$ and $3.16\times$ at $N = 3$ against a linear prediction of 2 and 3 (Nagarajan and Ashok, 2026). This
 116 is inherited scaffolding; everything below replaces one object in it.

117 **Two questions, and they are not the same question.** Is each agent stable alone, $m_1 < 1$? And
 118 is the collection stable when its members interact, $m_N < 1$? A shared environment separates
 119 them, because the loop closes through that environment and nothing in dealer A’s validation set
 120 contains dealer B. **Agent count is not a descriptive characteristic of a market; it enters the**
 121 **stability condition.** Read as economics, this is an externality, and it is *technological rather than*
 122 *pecuniary* (Buchanan and Stubblebine, 1962): firms affecting each other only through prices impose
 123 no market failure, since the price movement is a transfer, while firms that reshape the distribution
 124 their competitors learn from change the production technology of everyone’s forecast, which is a
 125 genuine efficiency loss. We call m_N the **feedback reproduction number**: the factor by which a
 126 common perturbation is amplified per retraining round.

127 **Objects and standing assumptions.** Firm i has a response Jacobian $E_i \in \mathbb{R}^{d \times d}$. The *alignment*
 128 *matrix* $R = (r_{ij})$ collects $r_{ij} = \langle \text{vec } E_i, \text{vec } E_j \rangle / (\|E_i\|_F \|E_j\|_F)$, a correlation matrix of feedback
 129 directions rather than of returns; it is positive semidefinite with unit diagonal.

130 **Assumption 1** (Standing). (A1) All spectral statements linearize the joint retraining map around
 131 the joint equilibrium, with the simulator as the nonlinear check. (A2) Theorem 2 is exact for equal
 132 moduli $m_i = m_1$; unequal moduli give a two-sided bound. (A3) Firms retrain on a common clock
 133 at a common cadence K . (A4) Firms couple through one pool with a scalar spillover κ . No sign
 134 condition on R is assumed; the one place one is needed is stated where it is used, in Section 7.

135 **The reduction.** The substitution of R for $\mathbf{1}\mathbf{1}^\top$ is the whole paper, so it is derived rather than
 136 posited.

137 **Lemma 1** (Reduction). Assume (H1) firm i ’s deviation from equilibrium is a scalar x_i along its unit
 138 response direction $\hat{E}_i = E_i / \|E_i\|_F$, so the state is $x \in \mathbb{R}^N$ rather than $\mathbb{R}^{Nd}$; (H2) $\|E_i\|_F = \varepsilon$ for

Table 1: Anchors of Theorem 2. The simplex is the sharper counterexample to mean-based diversity: it *minimizes* mean alignment yet exceeds orthogonality’s $\lambda_{\max}$.

Configuration	$\lambda_{\max}(R)$	N_{eff}	Reading
Monoculture, $R = \mathbf{1}\mathbf{1}^\top$	N	$1 + \kappa(N - 1)$	the inherited law
Orthogonal, $R = I$	1	1	a hundred firms are dynamically one firm
Simplex, $r_{ij} = -1/(N-1)$	$N/(N-1)$	$1 + \kappa/(N-1)$	maximal diversity, and the mode labels swap
Supply chain, share s	$1 + s(N - 1)$	$1 + \kappa s(N - 1)$	the number of independent <i>models</i>

every i , which is (A2) restated in the new object; (H3) firm i retrains against the component of the pool distortion along $\hat{E}_i$, feeling its own contribution in full and each competitor’s with weight κ ; and (H4) firm i minimizes a γ -strongly-convex objective with sensitivity β to the felt distortion, so $x_i^+ = -(\beta/\gamma) \text{felt}_i$. Then the joint retraining map is linear with Jacobian

$$J = -m_1[(1 - \kappa)I + \kappa R], \quad m_1 = \varepsilon\beta/\gamma. \quad (1)$$

Proof sketch. The pool carries the aggregate distortion $Z = \sum_j x_j E_j$, so by (H3) firm i feels $\text{felt}_i = \langle \hat{E}_i, x_i E_i \rangle + \kappa \sum_{j \neq i} \langle \hat{E}_i, x_j E_j \rangle$. By (H2), $\langle \hat{E}_i, E_j \rangle = \varepsilon r_{ij}$, and since $r_{ii} = 1$ the own term folds into the sum, leaving $\text{felt}_i = \varepsilon[(1 - \kappa)x_i + \kappa(Rx)_i]$. Applying (H4) coordinatewise gives (1), with m_1 the single-firm modulus of Perdomo et al. (2020). $\square$

Alignment enters through (H3): a firm senses the pool through its own response channel, so two firms perturbing the environment in orthogonal directions do not read each other at all, however large those perturbations. (H1) is the substantive hypothesis; without it J carries a Kronecker structure on $\mathbb{R}^{Nd}$ whose radius is not in general $N_{\text{eff}} m_1$, the block-secular problem deferred in Section 9.

4 Result 1: the effective number of independent learners

Counting firms is wrong, and the correction is where monoculture and supply-chain content enter. $\lambda_{\max}(R)$ runs from 1, all responses orthogonal, to N , monoculture.

Theorem 2 (Alignment sets the reproduction number). *Under Assumption 1 with equal moduli and heterogeneous response directions, the joint Jacobian is (1) and the market’s spectral radius is*

$$m_N = N_{\text{eff}} m_1, \quad N_{\text{eff}} = 1 + \kappa(\lambda_{\max}(R) - 1), \quad (2)$$

so the market is stable if and only if $m_N < 1$.

Setting $R = \mathbf{1}\mathbf{1}^\top$ recovers the symmetric multi-dealer law of Nagarajan and Ashok (2026) exactly, so the published result is not generalized away but located: it is the monoculture corner, where every firm’s feedback points the same way. Four anchors follow, each checked independently (Table 1). $N_{\text{eff}} \geq 1$ always, so interaction never stabilizes a market below what its members achieve alone.

The supply chain. Decompose $E_i = \sqrt{s} E_{\text{shared}} + \sqrt{1-s} \Xi_i$, where $s \in [0, 1]$ is the fraction attributable to a shared foundation model, vendor or pretraining corpus. Alignments concentrate at $r_{ij} \rightarrow s$ with $O(1/d)$ fluctuation and an explicit failure probability, so $N_{\text{eff}} \rightarrow 1 + \kappa s(N - 1)$ and every result below inherits s for free: fifty dealers fine-tuning one vendor’s model are dynamically $\sim 1 + 49\kappa$ learners at $s \approx 1$ and barely more than one at $s \approx 0$.

Why the naive diversity index fails, twice. Mean alignment understates *clustered* alignment: three tightly aligned firms among ten otherwise-orthogonal ones give $N_{\text{eff}} = 2.60$ at $\kappa = 0.8$ against 1.48 for the uniform configuration with the identical mean, so a market at $m_1 = 0.5$ runs $m_N = 1.30$ and is unstable while the mean index reports 0.74 and calls it safe with margin. The simplex is sharper: it minimizes mean alignment at $-1/(N - 1)$ yet its $\lambda_{\max} = N/(N - 1)$ exceeds orthogonality’s 1, so the mean *does not order configurations correctly* and fails as a ranking, not only as a level. The failure is signed: $N_{\text{eff}} \geq 1 + \kappa(N - 1)\bar{r}$ for every R , so the mean index always under-states risk.

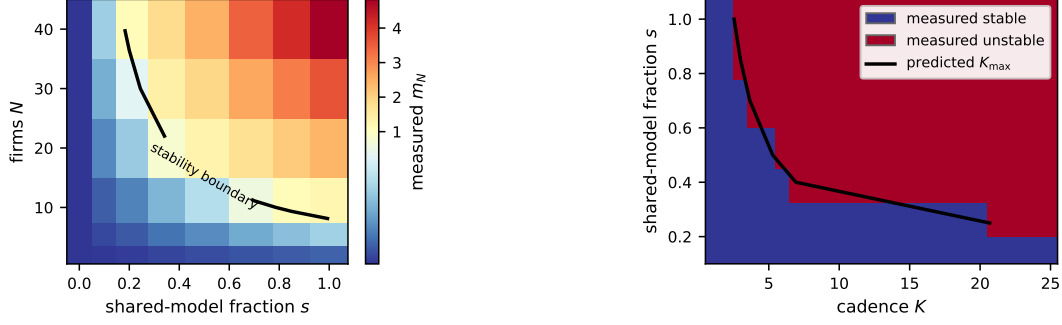


Figure 1: Left: the (N, s) phase diagram over 48 cells (Theorem 2). Right: the cadence window over 175 cells of the (N_{eff}, K) grid against predicted K_{max} , with s contours and $c = 0.8$ (Theorem 3); the realized window is $\lfloor K_{\text{max}} \rfloor$ and the plotted frontier continuous. Both are dry runs (Section 8).

5 Result 2: the crowding-cadence frontier

The first lever is quantity regulation, and it is the one a firm can pull alone: retrain less often. A learner taking only K gradient steps per deployment realizes outer-map slope $\mu(K) = -m + c^K(1 + m)$, where $c = 1 - \eta\gamma$ is the inner per-step contraction (Mendler-Dünner et al., 2020). Because c is built from the learner’s own objective curvature and step size, it does not depend on how many competitors it has, and that independence is what makes the composition below work. We verify it by measuring the realized contraction inside the joint market.

Theorem 3 (Systemic cadence window). *Under Assumption 1, in the N -firm market with synchronous K -step retraining the joint deployment map is $c^K I + (1 - c^K)J$, so the binding mode has slope $\mu_N(K) = -m_N + c^K(1 + m_N)$ and the market is stable if and only if*

$$K < K_{\text{max}} = \frac{\ln((m_N - 1)/(m_N + 1))}{\ln c}, \quad (3)$$

with K_{max} infinite when $m_N \leq 1$ and strictly decreasing in m_N .

The slope on the mode carrying $\lambda_i(R)$ is strictly decreasing in λ_i , which puts every slope at or below $c^K < 1$, so the upper side of the constraint never binds at any cadence, and places the binding mode at λ_{max} whatever the sign pattern of R .

Critical crowding. The same inequality reads $m_N < (1 + c^K)/(1 - c^K)$, and since K is a positive integer the most permissive case is $K = 1$. A stabilizing cadence exists at all if and only if $m_N < (1 + c)/(1 - c)$, a factor of 9 at $c = 0.8$. Past that the market is unstable at every retraining frequency. The laziest retrainer is therefore the hardest to destabilize.

What the supply chain does to the window. Substituting Result 1 makes K_{max} a function of vendor concentration at a fixed number of firms: at $m_1 = 0.15$, $\kappa = 0.8$, $N = 30$, $c = 0.8$, the window runs $K_{\text{max}} = 20.68$ at $s = 0.25$, 5.28 at $s = 0.5$ and 2.53 at $s = 1$. Holding competitors fixed and raising vendor concentration cuts every incumbent’s retraining budget by a factor of eight. The externality in its sharpest form is therefore not merely that **your competitor’s entry consumes your retraining budget** but that **your competitor’s choice of vendor does too**, without their entering at all.

What the lever costs. Cadence buys stability with staleness, and the exchange rate is the same quantity on both sides: after a retraining round the deployed model’s gap to its own best response is exactly c^K times what it was, so the c^K that widens the stability margin is the staleness.

6 Result 3: herd immunity, and diversity as its substitute

The second lever is a technology mandate, and unlike cadence it is not something a firm can pull alone to any effect. A *corrected* firm runs the feedback-aware update (Izzo et al., 2021; Miller et al.,

204 2021), so its retraining is governed by the objective curvature γ_{PO} in place of the cobweb and its
 205 modulus is scaled by $\theta = \gamma/\gamma_{PO} < 1$. What is new is running it inside the game: a mixed market of
 206 N_b blind firms and $N - N_b$ corrected ones, corrected fraction $\rho = 1 - N_b/N$. Correction changes a
 207 firm's gain and not its response direction, so R is untouched.

208 **Theorem 4** (Mixed-market stability). *Under Assumption 1, with supply-chain alignment and two*
 209 *correction levels, the joint Jacobian's symmetric congruence is a diagonal matrix plus a positive*
 210 *rank-one term, so its spectrum solves a secular equation. Two distinct moduli collapse it to a*
 211 *quadratic, and for $1 \leq N_b \leq N - 1$ the radius is its larger root,*

$$\rho(J) = \frac{1}{2}(P + \sqrt{P^2 - 4Q}), \quad P = (1 - \tilde{\kappa})m_1(1 + \theta) + \tilde{\kappa}m_1(N_b + \theta N_c), \quad Q = \theta m_1^2(1 - \tilde{\kappa})N_{\text{eff}}, \quad (4)$$

212 with $\tilde{\kappa} = \kappa s$ and $N_c = N - N_b$. An empty block needs the single-block form instead, the quadratic
 213 otherwise leaving a phantom root that can exceed the true radius.

214 N_{eff} appearing inside Q is why every statement below carries s : the mixed market inherits Result 1's
 215 crowding through the product term, not by stipulation.

216 **The strong-correction limit, and it errs in the unsafe direction.** As $\gamma_{PO} \rightarrow \infty$ the constant
 217 term vanishes and the unstable cycle runs through blind firms only, giving $N_b < N_c(s) = 1 +$
 218 $(1/m_1 - 1)/(\kappa s)$. That criterion taken alone is **optimistic rather than conservative**. The radius is
 219 nondecreasing in θ by Perron-Frobenius on an entrywise-nonnegative matrix, so the limit under-states
 220 it: on random draws it calls 11.8% of configurations stable that are unstable at every finite correction
 221 strength. The exact root is therefore the criterion and the limit its corollary, quoted with its error
 222 direction. The same monotonicity carries the reassuring half: **correction never backfires**, so no
 223 configuration exists in which un-blinding a firm destabilizes the market.

224 **The collapse, and its honest generalization.** At $\kappa = s = 1$ the limit reads $\rho^* = 1 - 1/m_N$,
 225 **exactly the epidemiological herd-immunity threshold** $1 - 1/R_0$ with the systemic modulus as the
 226 reproduction number. A market of 10 firms at $m_N = 2.5$ needs 60% of them un-blinded *if correction*
 227 *is perfect*. It is not, and at that same corner the quadratic degenerates and the stability condition
 228 solves exactly for

$$\rho^*(e) = \frac{1 - 1/m_N}{e}, \quad e = 1 - \gamma/\gamma_{PO}, \quad (5)$$

229 with e the **correction efficacy**. That is the standard coverage requirement for an *imperfect vaccine*,
 230 of which the clean law is the perfect-efficacy corner. The clean form assumes homogeneous mixing
 231 (Fine et al., 2011), and the collapse here assumes the $\kappa = s = 1$ corner for precisely that reason.

232 **Corollary 5** (Critical efficacy). $\rho^*(e)$ exceeds one, so no corrected fraction works at all, exactly
 233 when $\theta > 1/m_N$. Correction is a usable lever only if $\gamma_{PO}/\gamma > m_N$.

234 At $m_N = 2.5$ the corrected update must deliver a $2.5\times$ curvature improvement or un-blinding cannot
 235 stabilize the market whatever fraction is treated. This parallels Section 5's critical crowding: each
 236 lever has a regime past which it stops working, and knowing both is what makes trading them off
 237 honest rather than an extrapolation.

238 **The synthesis.** ρ^* is increasing in s , strictly wherever positive, so a market reaches stability along
 239 either axis. At $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$: a monoculture ($s = 1$) needs $\rho^* = 0.596$, that is 12
 240 corrected firms; at $s = 0.5$, $\rho^* = 0.242$, five firms; at $s = 0.2$, $\rho^* = 0$, so the market is stable with
 241 no corrected agents at all. The regulator's count is $N - \lceil N_c \rceil + 1$ rather than $\lceil \rho^* N \rceil$, and the two
 242 differ by a firm exactly when N_c is an integer.

243 **The economic reading.** Correction is a public good: an un-blinded firm captures a private benefit
 244 while the stability it contributes accrues to everyone, so the market free-rides below the threshold
 245 and does not reach ρ^* unaided (Bergstrom et al., 1986). Model diversity has the same structure, a
 246 firm choosing an idiosyncratic vendor paying privately and supplying systemic stability for free. **The**
 247 **substitution is therefore between two goods that are both under-supplied**, not between a good
 248 and a bad.

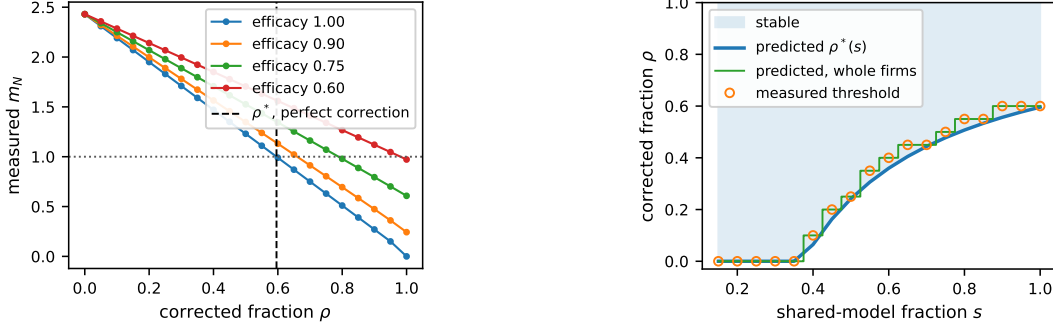


Figure 2: Left: stability against corrected fraction at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$, $s = 1$, across efficacies; the gap between the exact threshold and the perfect-correction limit is a result, not an error bar. Right: **the substitution frontier**, the (ρ, s) iso-stability curve, matching at 18 of 18 points. Both are dry runs (Section 8).

7 Result 4: the Pigouvian wedge

The third lever is the price instrument. Inside the stable region the binding mode is an AR(1) with coefficient $-m_N$ driven by noise of variance σ^2 , so its stationary variance is $V(m_N) = \sigma^2/(1-m_N^2)$: the smooth proxy for divergence risk, divergent exactly at the boundary the rest of the paper is about, and nothing below uses more than V being positive, increasing and convex. The sign the variance does not see is not lost: the lag-1 autocorrelation is $-m_N$ and therefore *negative*, so a crowded market's common mode oscillates rather than persists, which is the observable the supervision estimator of Section 9 reads.

Each firm picks an adaptation aggressiveness a_i , meaning cadence, learning rate or responsiveness, buying concave tracking benefit $B(a_i)$ through $m_i = \mu(a_i)$. Firm i bears weight w_i on common-mode variance and the planner W , with $W > \sum_i w_i$ because every trade has a client bearing execution-quality variance that no dealer's objective contains. That is an assumption, and the results turn on its *sign* and never its size.

Lemma 6 (Marginal crowding share). *At the symmetric point, $\partial m_N / \partial m_i = N_{\text{eff}} v_i^2$ where v is the leading eigenvector of the coupling matrix. The shares sum to N_{eff} rather than to one, and on exchangeable R each equals N_{eff}/N .*

Where a sign condition is needed, and where it is not. Theorem 2's radius needs none: R is positive semidefinite with unit diagonal, so every $\lambda_i(R) \geq 0$ and $1 + \kappa(\lambda_i - 1)$ increases in λ_i , placing the radius at $\lambda_{\max}$ whatever the off-diagonal signs. What needs one is the *reading* of Lemma 6, that the binding mode is the common one so v is a vector of market shares. **(A5)** supplies it: with R entrywise nonnegative, Perron-Frobenius places the leading eigenvector in the nonnegative orthant, and shared vendors and corpora produce positive alignment.

Both readings of Lemma 6 matter. The *sum* is N_{eff} , so the market's total sensitivity to a uniform increase in aggressiveness is amplified by exactly the effective learner count of Result 1. The *individual* share is N_{eff}/N , the arithmetic shape of every commons problem. The naive guess here is $1/N$ by analogy with the commons, and it is wrong by the factor N_{eff} , which at $N = 20$, $\kappa = 0.8$, $s = 1$ is 16.2.

Theorem 7 (The wedge). *Under Assumption 1 and the welfare assumptions above, at a symmetric profile the private and social first-order conditions are the same equation with a different price on crowding, and the marginal cost a firm ignores is*

$$t^* = (W - w_i) V'(m_N) \frac{N_{\text{eff}}}{N} \mu'(a), \quad V'(m_N) = \frac{2\sigma^2 m_N}{(1 - m_N^2)^2}. \quad (6)$$

A per-unit fee t^* on aggressiveness makes the private condition coincide with the social one, so the taxed decentralized equilibrium implements the symmetric social optimum.

With symmetric weights the private condition ignores the fraction $(N-1)/N$ of the marginal variance cost borne by other firms, plus the entire client-side exposure. t^* is strictly increasing in N , κ , s and

Table 2: Panel coverage and status. Panel 1 is external validation against a prior published run, not a self-consistency check.

Panel	Tests	Status	Outcome
1. Amplification	Section 3	measured	$1.7428\times, 3.1567\times$; relative error 0.00
2. (N, s) phase diagram	Thm 2	dry run	48 cells, max error 7.1×10^{-15}
2b. Clustered companion	Thm 2	dry run	measured m_N 1.30; mean index says 0.74
3. Cadence frontier	Thm 3	dry run	175 cells, zero disagreements
4. Herd immunity	Thm 4	dry run	thresholds 12/14/16/20 firms across efficacy
5. Substitution frontier	Thm 4	dry run	18 of 18 points exact
6. Over-adaptation	Thm 7	dry run	12 configurations, zero contradicting Cor. 8

m_N , and diverges at the boundary like $(1 - m_N)^{-2}$. The rate is the economically loaded part: a regulator setting a fee from a comfortable-looking market and holding it fixed is setting it far too low by the time the market is close to instability.

The provenance channel. m_N depends on s , so the wedge has a second channel: $\partial m_N / \partial a_i = (N_{\text{eff}}/N)\mu'(a_i)$ against $\partial m_N / \partial s = m_1\kappa(N - 1)$. The same wedge therefore prices **shared-model adoption** directly, turning the monoculture externality into a fee rather than a warning, and the two channels are asymmetric. The aggressiveness factor N_{eff}/N is bounded and tends to κ from above, so a firm's marginal footprint through its own cadence does not grow without bound, while the provenance channel is **linear in N and does not decay** against a private adoption gain that does not grow with market size at all. Unregulated adoption therefore runs toward the unstable configuration, which is why the diversity floor of Section 6 is a different instrument from the cadence cap of Section 5 rather than a restatement of it.

Corollary 8 (Over-adaptation). *The decentralized equilibrium over-adapts relative to the social optimum for every $N \geq 2$, strictly so even with client exposure switched off, and the distortion grows as the market crowds. At $N = 1$ with no client exposure the wedge is exactly zero.*

The fraction the private condition ignores is free of N_{eff} : with no client exposure it is exactly $(N - 1)/N$ whatever κ and s are. Crowding enters the wedge's *level*, through $V'(m_N)$, and not the distortion's *shape*, so Corollary 8 is the generic commons result and the provenance asymmetry above is the market-of-models one. Sections 5, 6 and 7 are respectively quantity regulation, a technology mandate or structural remedy, and the price instrument; choosing among them under uncertainty is Weitzman's question (Weitzman, 1974), and we supply the objects needed to pose it rather than answering it.

8 Experiments

Six panels, one per result, each with its evidential status in its own row of Table 2, because the six are not the same kind of evidence. A panel marked **measured** ran in the order-flow simulator, a genuine N -dealer market with informed flow, spread and inventory that reduces bit for bit to the single-dealer market at $N = 1$. A panel marked **dry run** ran in a linearized reference environment realizing the model's response geometry and nothing else: it checks that the closed forms were derived from the map we believe they were derived from, which is a check on the derivation rather than evidence about dynamics, and it has caught real errors. It establishes nothing about a market. Stability everywhere is estimated from the trajectory's asymptotic growth rate under the actual retraining map rather than read off an eigensolve. All runs are CPU-only and deterministic from (config, seed), with common random numbers on paired probes; every closed form carries a numerical certificate and the harness exits nonzero on any disagreement. The simulator is described generically pending an artifact release; proofs and specifications are in the appendix, code to reviewers as anonymized supplementary material.

The one measured panel. Panel 1 reproduces a prior published run in that run's own simulator, at the relative error Table 2 records. **The gap to the linear prediction is content, not error:** the market runs -12.9% at $N = 2$ and $+5.2\%$ at $N = 3$, with *opposite signs*, which is what a nonlinear market with saturating flow does around a linearization. The reference environment reproduces

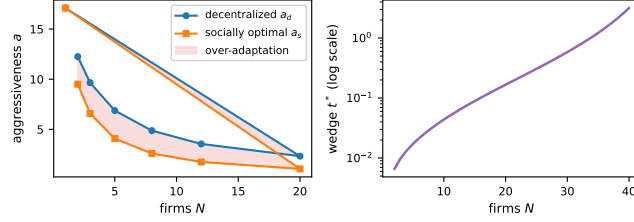


Figure 3: Over-adaptation and the wedge. Left: decentralized against socially optimal aggressiveness across N at $\chi = 1$, $s = 1$. Right: t^* against N , log scale. Dry run; the boundary divergence fits $(1 - m_N)^{-2}$ at log-log slope -2.000 .

the prediction exactly because it omits precisely that nonlinearity. The differential mode measures 3.4×10^{-3} against a theoretical zero at $\kappa = 1$, so the instability is purely common-mode, the mechanism Theorem 2 generalizes.

Why the rest are not measured. A heterogeneous-response port of the order-flow simulator was built and is exact at flat exposure profiles, reproducing the unmodified simulator at every N from 1 to 6, but its acceptance gate failed: the liquidity and price-impact channels stay shared whatever the response profiles are, a residual reaching 17.7 percentage points at $N = 2$, larger than several effects these panels resolve, and at $N \geq 4$ the finite-difference probe no longer measures a local slope. We drew no figure from that port and moved no status. Panel 6 is a dry run for a different reason: the order-flow simulator carries no aggressiveness choice variable and no welfare object, so no measured counterpart exists to be built.

In panel 6 both first-order conditions are solved by bisection with m_N read from realized dynamics rather than the closed form, so the over-adaptation verdict is a property of the simulated market rather than of the algebra: the gap widens from 29.0% at $N = 2$ to 119.9% at $N = 20$, and the degenerate $N = 1$ case sits at exactly zero, as Corollary 8 requires.

9 Limitations and conclusion

Supervision from public prices. Every quantity above is hidden from the regulator these results address, and near the boundary the system tells on itself: as m_N approaches one, the leading principal component of public quotes has lag-1 autocorrelation approaching one in magnitude, negative because the common mode oscillates, and variance share growing like $1/(1 - m_N)$. A supervisor can therefore estimate distance to instability from public prices alone, and the estimate sharpens as the market nears the boundary it watches for. Its consistency proof, sample complexity and real-data panel are deferred, the last scoped in the appendix as consistency evidence with a placebo, never identification.

Limitations. The analysis is linearized around the joint equilibrium (A1), and panel 1 shows what that costs in a real market. **The heterogeneous sweep is closed-form agreement, not measurement,** the largest gap in the work; routing the liquidity and impact channels through the response profiles is the identified repair. Theorem 2 is exact for equal moduli (A2), unequal ones giving a two-sided bound, and share-weighted alignment, the block-secular reduction when (H1) fails, and the sharp concentration rate are named extensions. The $1 - 1/R_0$ collapse assumes the $\kappa = s = 1$ corner, three or more correction levels return a genuine secular equation, and correction moving a response direction rather than a gain is not modelled. Deployment is synchronous (A3); heterogeneous clocks break the eigenvector sharing Theorem 3 uses. Theorem 7 prices instability through stationary variance, a modeling choice. Neither R nor s is observable to an outsider, which the estimator above answers with its real-data leg deferred.

Conclusion. Individual model evaluation cannot certify a market of models. The quantity separating the two questions is the effective number of independent learners, which no evaluation of a single model can see: a firm can pass every stability test available to it and still be one of the reasons its market fails. Regulation of shared general-purpose models has thresholds on model scale but no mechanism tying model sharing to a measurable systemic quantity. This paper supplies one, with two substitutable instruments and a price.

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A Notation and standing assumptions

Symbols marked *new* are this paper’s contribution surface. Everything else is inherited from the base performative-retraining model and cited rather than re-derived.

Symbol	Meaning	Range
N	firms in the market	integer ≥ 1
d	dimension of each firm’s decision	integer ≥ 1
ε	performative response strength of the environment	> 0
β	sensitivity of the objective to the induced shift	> 0
γ	own-objective curvature	> 0
m_1	single-firm retraining modulus, $\varepsilon\beta/\gamma$	stable iff < 1
κ	spillover between firms through the shared pool	$[0, 1]$
E_i	firm i ’s response Jacobian (<i>new</i>)	$d \times d$
r_{ij}	pairwise alignment of feedback directions (<i>new</i>)	$[-1, 1]$
R	alignment matrix (r_{ij}) (<i>new</i>)	PSD, unit diagonal
$\lambda_{\max}(R)$	effective number of independent learners (<i>new</i>)	$[1, N]$
N_{eff}	effective crowding, $1 + \kappa(\lambda_{\max}(R) - 1)$ (<i>new</i>)	$[1, 1 + \kappa(N - 1)]$
m_N	systemic modulus, $N_{\text{eff}}m_1$ (<i>new</i>)	stable iff < 1
s	shared-model fraction (<i>new</i>)	$[0, 1]$
K	gradient steps per deployment	integer ≥ 1
η	inner learning rate	> 0
c	inner per-step contraction, $1 - \eta\gamma$	$(0, 1)$
γ_{PO}	curvature governing the corrected dynamics	$> \gamma$
θ	γ/γ_{PO} , the correction factor (<i>new</i>)	$(0, 1]$
e	correction efficacy, $1 - \theta$ (<i>new</i>)	$[0, 1]$
N_b	blind (uncorrected) firms (<i>new</i>)	$0 \dots N$
ρ^*	herd-immunity threshold (<i>new</i>)	$[0, 1)$
a_i	firm i ’s adaptation aggressiveness (<i>new</i>)	> 0
t^*	the Pigouvian wedge (<i>new</i>)	

The standing assumptions of Assumption 1 are (A1) linearization of the joint retraining map around the joint equilibrium, with the simulator as the nonlinear check; (A2) equal moduli $m_i = m_1$, under which Theorem 2 is an identity and outside which it is the two-sided bound of Proposition B.6 below; (A3) synchronous retraining on a common clock at a common cadence K ; and (A4) coupling through one pool with a scalar spillover κ . The sign condition (A5), that R is entrywise nonnegative, is not a standing assumption: Theorem 2’s radius does not need it, and it is invoked only in Section 7, where the reading of Lemma 6 depends on it. This appendix makes both halves of that statement precise.

B The reduction and the alignment spectrum

This appendix proves Lemma 1 and Theorem 2, establishes the properties of R they use, gives the three anchors, and proves that mean-based diversity indices err with a sign.

B.1 Hypotheses and the reduction

Firm i deploys $h_i \in \mathbb{R}^d$, and its deployment reshapes the flow it faces through the response Jacobian $E_i \in \mathbb{R}^{d \times d}$. Write $\text{vec } E_i$ for the d^2 -vector form, $\|\cdot\|_F$ for the Frobenius norm, and $\hat{E}_i = E_i/\|E_i\|_F$ for firm i ’s unit response direction. The reduction uses four hypotheses.

(H1) Rank-one deviation. Firm i ’s deviation from the joint equilibrium is a scalar x_i along its own response direction, so the market state is $x \in \mathbb{R}^N$ rather than $\mathbb{R}^{Nd}$.

(H2) Equal response magnitude. $\|E_i\|_F = \varepsilon$ for every i . This is (A2) restated in the new object: the inherited response strength ε is the Frobenius norm of the response Jacobian.

(H3) Own-channel sensing. Firm i retrains against the component of the pool distortion lying along its own response direction, feeling its own contribution in full and each competitor’s with weight κ .

(H4) Quadratic retraining. Firm i minimizes a γ -strongly-convex own objective with sensitivity β to the felt distortion, giving $x_i^+ = -(\beta/\gamma)$ felt $_i$.

430 *Proof of Lemma 1.* The pool carries the aggregate distortion $Z = \sum_j x_j E_j$. By (H3) the distortion
 431 firm i feels is

$$\text{felt}_i = \langle \hat{E}_i, x_i E_i \rangle + \kappa \sum_{j \neq i} \langle \hat{E}_i, x_j E_j \rangle = \varepsilon \left[x_i + \kappa \sum_{j \neq i} r_{ij} x_j \right],$$

432 using $\langle \hat{E}_i, E_j \rangle = \varepsilon r_{ij}$, which is (H2). Since $r_{ii} = 1$ the own term folds into the sum,

$$\text{felt}_i = \varepsilon \left[(1 - \kappa) x_i + \kappa \sum_j r_{ij} x_j \right].$$

433 By (H4), $x^+ = -(\beta/\gamma)\varepsilon[(1 - \kappa)I + \kappa R]x$, and $m_1 = \varepsilon\beta/\gamma$ is the single-firm modulus, which is
 434 (1). $\square$

435 The alignment enters because firm i senses the pool through its own response channel, so two
 436 firms whose deployments perturb the environment in orthogonal directions do not read each other's
 437 perturbations at all, however large those perturbations are. Hypothesis (H3) is what carries that
 438 mechanism. If every E_i is equal then $r_{ij} = 1$ throughout, $R = \mathbf{1}\mathbf{1}^\top$, and the reduction returns the
 439 inherited shared-pool law exactly, so the base model is the corner of the new one in which every
 440 firm's feedback points the same way. Certificate P1 builds random E_i , forms the retraining map from
 441 the felt distortion without ever constructing R , differentiates it numerically and compares against
 442 (1), agreeing to 5.6×10^{-17} at (N, d, κ) equal to $(3, 4, 0.8)$, $(5, 6, 0.35)$, $(8, 5, 1.0)$ and $(6, 7, 0.0)$,
 443 so both spillover endpoints are covered.

444 Hypothesis (H1) is the substantive one. Without it the state is $(h_1, \dots, h_N) \in \mathbb{R}^{Nd}$ and the Jacobian
 445 carries a Kronecker structure with block (i, j) equal to $-(\beta/\gamma)c_{ij}E_j$, where $c_{ij} = 1$ for $i = j$
 446 and κ otherwise, whose spectral radius is not in general $m_1 N_{\text{eff}}$. Reducing that object is the fully
 447 heterogeneous block-secular problem, and it is deferred.

448 B.2 What R is

449 **Lemma B.1.** Let V be the $N \times d^2$ matrix whose rows are the unit vectors $\text{vec } \hat{E}_i$. Then $R = VV^\top$
 450 and: (1) R is positive semidefinite, so every $\lambda_i(R) \geq 0$; (2) $r_{ii} = 1$, hence $\text{tr } R = N$ and
 451 $\sum_i \lambda_i(R) = N$; (3) $|r_{ij}| \leq 1$; (4) $1 \leq \lambda_{\max}(R) \leq N$; (5) $\lambda_{\max}(R) = 1$ if and only if $R = I$; (6)
 452 $\lambda_{\max}(R) = N$ if and only if $R = D\mathbf{1}\mathbf{1}^\top D$ for a sign matrix $D = \text{diag}(\pm 1)$, that is, if and only if
 453 every response is parallel or antiparallel to one common direction.

454 *Proof.* (1) and (2) are immediate from $R = VV^\top$ with unit rows. (3) is Cauchy-Schwarz. For (4),
 455 the eigenvalues are nonnegative and sum to N , so the largest is at least the mean 1 and at most the
 456 total N . For (5), $\lambda_{\max} = 1$ with nonnegative eigenvalues summing to N forces all N eigenvalues to
 457 equal 1. For (6), $\lambda_{\max} = N$ forces every other eigenvalue to zero, so R has rank one, $R = vv^\top$, and
 458 unit diagonal gives $v_i^2 = 1$. $\square$

459 Item (4) is the one the paper leans on: $\lambda_{\max}(R)$ is a genuine effective count, bounded below by one
 460 firm and above by N , with both ends attained.

461 B.3 The spectrum

462 *Proof of Theorem 2.* Write $J = -m_1 B$ with $B = (1 - \kappa)I + \kappa R$. B is symmetric and shares
 463 its eigenvectors with R , with eigenvalues $\nu_i = (1 - \kappa) + \kappa\lambda_i(R)$, so $\rho(J) = m_1 \max_i |\nu_i|$. By
 464 Lemma B.1(1) every $\lambda_i(R) \geq 0$, and $\kappa \leq 1$ gives $1 - \kappa \geq 0$, so every $\nu_i \geq 1 - \kappa \geq 0$. The absolute
 465 value is therefore inert and the maximum sits at $\lambda_{\max}$ rather than at $\lambda_{\min}$:

$$\rho(J) = m_1 [(1 - \kappa) + \kappa\lambda_{\max}(R)] = m_1 [1 + \kappa(\lambda_{\max}(R) - 1)] = m_1 N_{\text{eff}}.$$

466 Stability of the linearized joint map is $\rho(J) < 1$; substituting $m_1 = \varepsilon\beta/\gamma$ and solving for ε gives the
 467 stated form. $\square$

468 Certificate P3 checks this on 384 random configurations with maximum deviation 4.7×10^{-15} .
 469 Dropping the absolute value is legitimate only because R is a Gram matrix. A symmetric unit-
 470 diagonal matrix with entries outside $[-1, 1]$ can put the radius on $\lambda_{\min}$ instead: at $N = 3$ with every

off-diagonal equal to -3 the spectrum is $\{-5, 4, 4\}$ and the negative mode dominates. Cauchy-Schwarz is what rules this out for a real alignment matrix, and P3 certifies both directions.

Corollary B.2. $N_{\text{eff}} \in [1, 1 + \kappa(N - 1)]$, attained at $R = I$ and $R = \mathbf{1}\mathbf{1}^\top$ respectively.

Corollary B.3. $\rho(J) \geq m_1$, with equality if and only if $R = I$. A market of adaptive models sharing an environment is never more stable than one of its members in isolation, and exact neutrality requires fully orthogonal responses.

Corollary B.3 says the externality has a sign: coupling can only amplify feedback, never damp it, so there is no configuration in which crowding does a market a favour. Individual modes can be more damped than m_1 , and the differential mode at $m_1(1 - \kappa)$ is one, but the radius governs and the radius cannot fall below m_1 .

B.4 Which mode binds

Proposition B.4. If R is entrywise nonnegative and irreducible then $B = (1 - \kappa)I + \kappa R$ is entrywise nonnegative and irreducible, so by Perron-Frobenius its leading eigenvector is strictly positive and simple, and the binding mode is a genuine common mode in which every firm deviates in the same direction.

Entrywise nonnegativity of R is exactly the condition that no firm's feedback anti-aligns with another's. Shared vendors and shared pretraining corpora produce positive alignment, so the realistic regime satisfies it, and the paper states the condition rather than assuming it silently. This is (A5), and Proposition B.4 is the only place it is used.

B.5 Three anchors, and a mode swap

Monoculture. $R = \mathbf{1}\mathbf{1}^\top$ has rank one with eigenvalue N on the all-ones direction, so $\lambda_{\max} = N$ and $N_{\text{eff}} = 1 + \kappa(N - 1)$. This is the inherited law whose predicted $2\times$ and $3\times$ at $N = 2, 3$ were measured at $1.74\times$ and $3.16\times$.

Orthogonal. $R = I$ gives $\lambda_{\max} = 1$, $N_{\text{eff}} = 1$ and $m_N = m_1$. A hundred firms perturbing the environment in a hundred orthogonal directions are dynamically one firm.

Simplex. With $r_{ij} = -1/(N - 1)$ off the diagonal,

$$R_{\text{simplex}} = \frac{N}{N-1}I - \frac{1}{N-1}\mathbf{1}\mathbf{1}^\top,$$

whose diagonal is $N/(N - 1) - 1/(N - 1) = 1$ as required. Since $\mathbf{1}\mathbf{1}^\top$ has eigenvalue N on the all-ones direction and 0 on its complement, the spectrum is 0 on $\mathbf{1}$ and $N/(N - 1)$ with multiplicity $N - 1$. So $\lambda_{\max} = N/(N - 1)$, $N_{\text{eff}} = 1 + \kappa/(N - 1)$, and the radius tends to m_1 as N grows. The value $m_1(1 - \kappa)$ does sit in the spectrum, carried by the all-ones direction at $\lambda = 0$, and it matches the inherited differential-mode eigenvalue derived by a different route, but it is not the spectral radius.

Under the simplex the all-ones direction carries the *smallest* eigenvalue, so the leading eigenvector is orthogonal to it and (A5) fails. That is consistent with Proposition B.4, since the simplex has negative entries and Perron-Frobenius does not apply. Certificate P4 checks all three parts, that the all-ones direction is annihilated, that the leading eigenvector is orthogonal to it, and that the entries are negative, at $N \in \{3, 5, 10, 30, 50\}$.

B.6 Mean-alignment indices err, and always in the same direction

Proposition B.5. Let $\bar{m}$ be the off-diagonal mean of R . Then $\lambda_{\max}(R) \geq 1 + (N - 1)\bar{m}$, hence $N_{\text{eff}} \geq 1 + \kappa(N - 1)\bar{m}$, with equality if and only if $\mathbf{1}$ is a leading eigenvector of R , that is, if and only if R has constant row sums.

Proof. $\mathbf{1}^\top R \mathbf{1} = N + N(N - 1)\bar{m}$, since the diagonal contributes N . The Rayleigh quotient at the all-ones vector gives $\lambda_{\max}(R) \geq \mathbf{1}^\top R \mathbf{1} / N = 1 + (N - 1)\bar{m}$, with equality if and only if $\mathbf{1}$ attains the Rayleigh maximum, that is if and only if $R\mathbf{1}$ is parallel to $\mathbf{1}$. Multiplying by κ and adding $1 - \kappa$ preserves the inequality. $\square$

515 Certificate P5 finds zero violations on 360 random R and confirms equality on uniform R . Two
516 readings follow. First, a mean-similarity diversity index never over-states systemic risk: it under-states
517 it or is exact, so every error it makes is in the unsafe direction. A regulator using mean similarity
518 is not merely imprecise, it is imprecise in the direction that lets unstable markets pass. Second,
519 $\mathbf{1}^\top R \mathbf{1} = \|\sum_i \text{vec } \hat{E}_i\|^2 \geq 0$ forces $\bar{m} \geq -1/(N-1)$, attained exactly at the simplex.

520 The gap is largest at the realistic topology, which is the vendor with a plurality rather than a monopoly.
521 Take $N = 10$ with three firms fully aligned and every other pair orthogonal, so that R is block
522 diagonal with a 3×3 all-ones block and $\lambda_{\max} = 3$. Three of the forty-five pairs are aligned, so
523 $\bar{m} = 1/15$ and the mean index reports $\lambda_{\max} = 1.6$. At $\kappa = 0.8$ the true N_{eff} is 2.60 against the
524 index's 1.48, an understatement by a factor of 1.757, and a market at $m_1 = 0.5$ then runs $m_N = 1.30$
525 and is unstable while the index reports 0.74 and calls it safe with margin.

526 The failure is not only one of level. The simplex *minimizes* the mean at $\bar{m} = -1/(N-1)$ while
527 carrying $\lambda_{\max} = N/(N-1)$, which exceeds one, so it is spectrally worse than orthogonality even
528 though it has the smaller mean. The mean therefore does not order configurations correctly, which is
529 why every stability claim in this paper uses the spectral form and never a mean.

530 B.7 Heterogeneous moduli

531 **Proposition B.6.** Let $M = \text{diag}(m_i)$ with every $m_i > 0$, and let $J = -M[(1-\kappa)I + \kappa R]$. Then

$$\max_i m_i \leq \rho(J) \leq \left(\max_i m_i\right) N_{\text{eff}}.$$

532 *Proof.* J is not symmetric, but the congruence $S = M^{1/2}$ gives $M^{1/2}(MB)M^{-1/2} =$
533 $M^{1/2}BM^{1/2}$, so $\rho(J) = \lambda_{\max}(A)$ with $A = (1-\kappa)M + \kappa M^{1/2}RM^{1/2}$, symmetric and pos-
534 itive semidefinite. For the lower bound, $e_i^\top A e_i = (1-\kappa)m_i + \kappa m_i r_{ii} = m_i$ since $r_{ii} = 1$, and the
535 Rayleigh quotient at each e_i gives $\lambda_{\max}(A) \geq m_i$ for every i . For the upper bound, $R \preceq \lambda_{\max}(R)I$
536 in the Loewner order and congruence by $M^{1/2}$ preserves it, so $M^{1/2}RM^{1/2} \preceq \lambda_{\max}(R)M$ and
537 hence $A \preceq N_{\text{eff}}M$, giving $\lambda_{\max}(A) \leq N_{\text{eff}} \max_i m_i$. $\square$

538 The bound is exact at $\kappa = 0$ and at $R = I$, where $A = M$ and both sides equal $\max_i m_i$, and its upper
539 half is exact at equal moduli, where it is Theorem 2. A mean-modulus formula is provably wrong:
540 at $R = I$ the firms decouple and $\rho(J) = \max_i m_i$, certified with m ranging over $\{0.1, \dots, 0.9\}$ at
541 $\rho = 0.900$ against a mean of 0.400. Certificate P8 finds no violation on 300 random draws.

542 C Concentration to the supply-chain limit

543 The decomposition $E_i = \sqrt{s} E_{\text{shared}} + \sqrt{1-s} \Xi_i$ is meant to deliver $R \rightarrow (1-s)I + s\mathbf{1}\mathbf{1}^\top$ and
544 therefore $N_{\text{eff}} \rightarrow 1 + \kappa s(N-1)$. Saying the cross terms vanish in expectation is not enough for a
545 stability claim, because a market is unstable or not at the realized alignment matrix rather than at its
546 mean.

547 **(D) Design assumption.** E_{shared} and the Ξ_i have independent entries, mean zero, unit variance,
548 sub-Gaussian with a common constant, and the Ξ_i are independent of each other and of
549 E_{shared} .

550 This is an idealization. Real response Jacobians are neither Gaussian nor independent across firms
551 beyond the shared component, and what the argument uses is sub-Gaussian entries and independence
552 of the idiosyncratic parts. Write $D = d^2$, $u = \text{vec } E_{\text{shared}}$ and $\xi_i = \text{vec } \Xi_i$, so that $\text{vec } E_i =$
553 $\sqrt{s}u + \sqrt{1-s}\xi_i$ in $\mathbb{R}^D$. Under (D), $\|u\|^2/D$ and $\|\xi_i\|^2/D$ concentrate at 1 while $u^\top \xi_j/D$ and
554 $\xi_i^\top \xi_j/D$ concentrate at 0, all at rate $1/\sqrt{D}$, so $r_{ij} \rightarrow s$ for $i \neq j$ and $\lambda_{\max}(R_\infty) = 1 + s(N-1)$
555 exactly by the rank-one structure.

556 **Lemma C.1.** Under (D) there is an absolute constant C such that for any $\delta \in (0, 1)$, with probability
557 at least $1 - \delta$,

$$\max_{i \neq j} |r_{ij} - s| \leq C \frac{\sqrt{\log(N/\delta)}}{d}.$$

558 *Proof.* Each of $\|u\|^2/D - 1$, $\|\xi_i\|^2/D - 1$, $u^\top \xi_i/D$ and $\xi_i^\top \xi_j/D$ is an average of D independent
559 mean-zero sub-exponential terms, so Bernstein's inequality gives a tail $2 \exp(-cDt^2)$ for $t \in (0, 1)$.
560 There are $O(N^2)$ such quantities across all pairs, and a union bound with $t = C\sqrt{\log(N/\delta)/D}$
561 controls all of them simultaneously with probability at least $1 - \delta$. The ratio defining r_{ij} is a smooth
562 function of these quantities with bounded derivatives in a neighbourhood of the limit point, so the
563 same rate carries to $r_{ij} - s$, and $\sqrt{D} = d$ gives the statement. $\square$

564 The inner products average over d^2 matrix entries, so the fluctuation is $1/d$. Certificate P7 measures
565 a log-log slope of -1.018 against a target of -1 , at $N = 8$, $s = 0.6$, $d \in \{8, 16, 32, 64, 128\}$ and 40
566 trials per point.

567 **Proposition C.2.** Under the event of Lemma C.1,

$$\begin{aligned} |\lambda_{\max}(R) - (1 + s(N - 1))| &\leq C \frac{N\sqrt{\log(N/\delta)}}{d}, \\ |N_{\text{eff}} - (1 + \kappa s(N - 1))| &\leq \kappa C \frac{N\sqrt{\log(N/\delta)}}{d}. \end{aligned}$$

568 *Proof.* Write $\Delta = R - R_\infty$, which has zero diagonal since both matrices have unit diagonal.
569 Weyl's inequality gives $|\lambda_{\max}(R) - \lambda_{\max}(R_\infty)| \leq \|\Delta\|_2$, and $\|\Delta\|_2 \leq \|\Delta\|_F \leq$
570 $\sqrt{N(N-1)} \max_{i \neq j} |\Delta_{ij}| \leq N \max_{i \neq j} |r_{ij} - s|$. Apply Lemma C.1; the N_{eff} form follows on
571 multiplying by κ . $\square$

572 The bound matters in relative terms, because N_{eff} itself grows with N . For large N the limit is
573 $1 + s(N - 1) \approx sN$, so the relative error is of order $C\sqrt{\log N}/(sd)$, with the N in the numerator
574 cancelling against the N in N_{eff} . The closed form is therefore accurate once $d \gg \sqrt{\log N}/s$, a
575 condition on dimension against vendor concentration rather than against firm count, and it degrades
576 as $s \rightarrow 0$, which is correct: at low shared-model fraction the limit matrix is close to the identity and
577 the leading eigenvalue is set by fluctuations rather than by the shared component. In the instantiation
578 the base model calibrates, d is in the low hundreds and N in the tens, so the requirement is met with
579 room for s above roughly 0.2.

580 The step $\|\Delta\|_2 \leq \|\Delta\|_F$ is crude and is used because it is elementary and unconditional. For a Δ
581 with the Wishart-type structure that (D) produces, the operator norm concentrates at $\sqrt{N}/d$ rather
582 than at N/d , which would improve the relative error to $1/(sd\sqrt{N})$, better for larger markets rather
583 than worse. The sharp rate is reported as measured and not as proved, since claiming it otherwise
584 would violate the paper's rule on status flags.

585 D The cadence composition

586 **Lemma D.1.** Let firm i retrain by gradient descent on a γ -strongly-convex quadratic own objective
587 whose minimizer is the frozen best response b , with step size η . Then K steps from z_0 land at
588 $z_K = b + c^K(z_0 - b)$ with $c = 1 - \eta\gamma$.

589 *Proof.* The gradient is $\gamma(z - b)$, so the update is $z \mapsto z - \eta\gamma(z - b) = b + (1 - \eta\gamma)(z - b)$. Induct
590 on K . $\square$

591 Here $c \in (0, 1)$ requires $0 < \eta < 2/\gamma$, and $\eta \leq 1/\gamma$ gives $c \in [0, 1)$. Certificate Q1 runs
592 actual gradient descent and measures the realized gap ratio, exact to 10^{-12} at three (η, γ) pairs and
593 $K \in \{1, 3, 7, 20\}$. The constant c is never written into the loop; it is whatever the loop does.

594 **Lemma D.2.** For a single firm whose full-retraining best response has slope $-m$, the K -step outer
595 map has slope $\mu(K) = -m + c^K(1 + m)$.

596 *Proof.* The frozen best response at the current deployment h_t is $b = -mh_t$, so by Lemma D.1,
597 $h_{t+1} = -mh_t + c^K(h_t + mh_t) = h_t[-m + c^K(1 + m)]$. $\square$

At $K \rightarrow \infty$ this gives $-m$, the full-retraining slope, and at $K = 0$ it gives the identity, since a firm taking no gradient steps does not move. The inherited single-firm lemma is therefore reproduced rather than assumed.

The composition rests on one hypothesis, stated so that it can be checked.

(C) The contraction c is the same scalar in the joint market as in the single-firm market. It is built from own-objective curvature γ and inner step size η , neither of which involves N , κ or R : competitors change *where* firm i is heading, through the frozen best response, and not *how fast* it gets there.

This is the only place the composition can fail, so it is certified rather than argued. Certificate C12 measures the realized one-step contraction inside the joint market across 45 configurations spanning $N \in \{1, 2, 5, 12, 30\}$, $\kappa \in \{0, 0.5, 1\}$ and $m_1 \in \{0.05, 0.15, 0.6\}$ with random R at each, and the spread across all of them is 4.4×10^{-16} .

Proof of Theorem 3. All firms compute their frozen best responses from the common current state x , giving $b = Jx$, then each takes K inner steps. By Lemma D.1 applied coordinatewise, and using (C) so that the same c applies in every coordinate,

$$x_{t+1} = Jx_t + c^K(x_t - Jx_t) = [c^K I + (1 - c^K)J]x_t.$$

This matrix is an affine function of J , so it shares J 's eigenvectors, with eigenvalues $\mu_i(K) = c^K - (1 - c^K)m_1\nu_i$ where $\nu_i = (1 - \kappa) + \kappa\lambda_i(R)$. On the mode carrying $\lambda_{\max}(R)$, where $m_1\nu_{\max} = m_N$ by Theorem 2, this is $\mu_N(K) = c^K - (1 - c^K)m_N = -m_N + c^K(1 + m_N)$.

For the stability claim, $\mu_i(K)$ is strictly decreasing in ν_i because $(1 - c^K)m_1 > 0$ for every $K \geq 1$. Every $\nu_i \geq 0$ by Lemma B.1(1) with $\kappa \leq 1$, so $\mu_i(K) \leq c^K < 1$ for all i and the upper side never binds, whatever R is; and $\min_i \mu_i(K)$ is attained at $\nu_{\max}$, so the constraint is exactly $\mu_N(K) > -1$. Expanding,

$$c^K - (1 - c^K)m_N > -1 \iff m_N < \frac{1 + c^K}{1 - c^K} \iff c^K > \frac{m_N - 1}{m_N + 1},$$

and taking logarithms, with $\ln c < 0$ flipping the inequality on division, gives $K < K_{\max}$. When $m_N \leq 1$ the right-hand side of the last inequality is non-positive while $c^K > 0$, so the constraint is vacuous and every cadence is stable. $\square$

Certificate Q2 builds the joint map by running the actual inner gradient descent on each basis vector and comparing against $c^K I + (1 - c^K)J$ at 32 combinations of (N, κ, m_1, K) , agreeing below 10^{-12} ; C8 checks the frontier on 51600 integer cases with zero mismatches; and Q5 iterates the actual joint map to convergence or divergence on 59 decisive trials at random (N, κ, s, m_1, K) , with zero disagreements with the predicted side.

The monotonicity argument replaces an earlier route through the size of the differential modes. It is shorter and it is unconditional: it needs no Perron-Frobenius condition, because monotonicity places the extreme at $\lambda_{\max}$ regardless of the sign pattern of R .

Corollary D.3. $K_{\max}$ is strictly decreasing in m_N , and hence in s .

Proof. With $f(m) = \ln \frac{m-1}{m+1}$ we have $f'(m) = \frac{1}{m-1} - \frac{1}{m+1} = \frac{2}{m^2-1} > 0$ for $m > 1$, and dividing by $\ln c < 0$ flips the sign. Since $m_N = m_1(1 + \kappa s(N-1))$ is increasing in s , the composition is decreasing in s . $\square$

Two readings the body uses. Since K is a positive integer, the realized window is $\lfloor K_{\max} \rfloor$ while the plotted frontier is continuous, and certificate Q4 confirms that $\lfloor K_{\max} \rfloor$ is the largest stable integer and $\lfloor K_{\max} \rfloor + 1$ is unstable at five (c, m_N) pairs. And a feasible cadence exists if and only if $K_{\max} > 1$, which is exactly $m_N < (1 + c)/(1 - c)$: past that level of effective crowding the market is unstable at every retraining frequency, and minimum-cadence operation multiplies the sustainable crowding by exactly $(1 + c)/(1 - c)$, a factor of 9 at $c = 0.8$. Critical crowding is therefore not a separate result but the $K = 1$ instance of the frontier.

Worked at $m_1 = 0.15$, $\kappa = 0.8$, $N = 30$ and $c = 0.8$, so that $N_{\text{eff}} = 1 + 23.2s$:

s	N_{eff}	m_N	K_{max}
0.25	6.80	1.020	20.68
0.50	12.60	1.890	5.28
1.00	24.20	3.630	2.53

All three m_N values sit below the critical crowding level of 9, which is why every column has a finite window. The table is not reproducible without c , so $c = 0.8$ is stated wherever it appears. The operational reading is that the externality is not only that a competitor's entry consumes a firm's retraining budget, but that a competitor's choice of vendor does too, without their entering at all and without either firm doing anything wrong.

Two scope notes close this appendix. By Lemma D.1 the deployed model's gap to the frozen best response after a round is exactly c^K times the gap before it, so c^K is simultaneously what buys stability and what measures the staleness of the deployed model: the lever is not free, and its price is quantified in the same parameter that delivers its benefit. And at $m_N = 1$ exactly, $\mu_N(K) = -1 + 2c^K$ lies strictly inside $(-1, 1)$ for every finite K , so lazy retraining strictly stabilizes the marginally unstable market, with a margin $2c^K$ that decays geometrically and falls below double precision past $K = 171$ at $c = 0.8$. That last is a floating-point limit and not a statement about markets, and the certificate records it as such.

E The mixed market

Of N firms sharing one pool, N_b are blind and run the ordinary retraining loop while $N_{\text{corr}} = N - N_b$ are corrected and run the feedback-aware update governed by γ_{PO} . Write $\theta = \gamma/\gamma_{PO} \in (0, 1]$, so that $\theta = 1$ is no correction and $\theta \rightarrow 0$ is the strong-correction limit.

(M) Correction scales a firm's modulus: a corrected firm carries θm_1 where a blind firm carries m_1 . This is the faithful reading of the single-firm result, in which the corrected loop's modulus is $m_1\gamma/\gamma_{PO}$. Correction changes a firm's gain and not its response direction, so R is untouched.

Take the supply-chain alignment $R = (1 - s)I + s\mathbf{1}\mathbf{1}^\top$ and write $k_s = \kappa s$, so that $B = (1 - k_s)I + k_s\mathbf{1}\mathbf{1}^\top$. By Proposition B.6 the radius of $J = -MB$ is $\lambda_{\max}(A)$ for $A = M^{1/2}BM^{1/2}$, and substituting B ,

$$A = (1 - k_s)M + k_s w w^\top, \quad w = M^{1/2}\mathbf{1}, \quad w_i = \sqrt{m_i}.$$

So A is a diagonal matrix plus a positive-weight rank-one term, and its eigenvalues not equal to a diagonal entry solve the secular equation $1 + k_s \sum_i m_i / ((1 - k_s)m_i - \lambda) = 0$. With only two distinct moduli the sum collapses to two terms and the equation clears to a quadratic, which is why the mixed market has a closed form at all and why three or more correction levels would not.

Proof of Theorem 4. Write $a = (1 - k_s)m_1$ and $b = (1 - k_s)\theta m_1$ for the two diagonal values. Clearing denominators in the secular equation gives

$$(a - \lambda)(b - \lambda) + k_s N_b m_1 (b - \lambda) + k_s N_{\text{corr}} \theta m_1 (a - \lambda) = 0,$$

that is $\lambda^2 - P\lambda + Q = 0$ with

$$P = a + b + k_s m_1 (N_b + \theta N_{\text{corr}}), \quad Q = \theta m_1^2 (1 - k_s) (1 + k_s (N - 1)) = \theta m_1^2 (1 - k_s) N_{\text{eff}}.$$

The two secular roots are this quadratic's roots, and the eigenvectors constant within each block and orthogonal to w supply the rest of the spectrum, namely a with multiplicity $N_b - 1$ and b with multiplicity $N_{\text{corr}} - 1$. Because the rank-one weight k_s is positive, the eigenvalues of A interlace those of $(1 - k_s)M$, so the largest eigenvalue of A exceeds every diagonal entry and is therefore the larger secular root rather than a degenerate one, giving $\rho(J) = \frac{1}{2}(P + \sqrt{P^2 - 4Q})$. $\square$

Certificate H1 reports a maximum error of 2.5×10^{-14} against dense eigensolves over 6000 random draws. The appearance of N_{eff} inside Q is not cosmetic: the mixed market inherits Theorem 2's crowding through that product, which is why every statement here carries s .

Degenerate blocks. When $N_b = 0$ or $N_b = N$ one block is empty, but the quadratic still carries that block's factor and leaves a phantom root behind. At $N_b = 0$ the naive quadratic returns $(1 - k_s)m_1$, a value no firm's modulus supports, and at small θ that phantom exceeds the true radius: 0.132 against a true 0.043 at $N = 12$, $m_1 = 0.3$, $\kappa = 0.8$, $s = 0.7$ and $\theta = 0.02$. The correct values are the single-block ones, $\rho = m_1 N_{\text{eff}}$ at $N_b = N$ and $\rho = \theta m_1 N_{\text{eff}}$ at $N_b = 0$, and any implementation must branch on this. Certified in H2.

Both limits are corollaries. At $\theta = 1$, $P = m_1[2(1 - k_s) + k_s N]$ and $Q = m_1^2(1 - k_s)N_{\text{eff}}$, whose roots are $m_1 N_{\text{eff}}$ and $m_1(1 - k_s)$, recovering Theorem 2 exactly. As $\theta \rightarrow 0$, $Q \rightarrow 0$, so one root goes to zero and the other to $P \rightarrow m_1(1 + \kappa s(N_b - 1))$, the blind-block condition. The argument that corrected firms transmit no feedback and drop out of the cycle is right, and it is now a corollary of the exact root rather than a separate claim.

E.1 The limit errs, and it errs optimistic

Proposition E.1. $\rho(J)$ is nondecreasing in θ .

Proof. $A_{ij} = \sqrt{m_i m_j} B_{ij}$ and B is entrywise nonnegative for $k_s \in [0, 1]$, so A is entrywise nonnegative. Every m_i is nondecreasing in θ , constant for blind firms and θm_1 for corrected ones, so every entry of A is nondecreasing in θ . Since A is symmetric and nonnegative, by Perron-Frobenius its Rayleigh maximizer may be taken in the nonnegative orthant, and for such an x the form $x^\top A(\theta)x$ is nondecreasing in θ . Taking x to be the maximizer at $\theta_1 < \theta_2$,

$$\lambda_{\max}(A(\theta_2)) \geq x^\top A(\theta_2)x \geq x^\top A(\theta_1)x = \lambda_{\max}(A(\theta_1)). \quad \square$$

Two readings follow and they point opposite ways. The reassuring one is that correction never backfires: more correction is always weakly stabilizing, so there is no perverse configuration in which un-blinding a firm destabilizes the market. The uncomfortable one is that $\rho(\theta) \geq \rho(0)$, so the strong-correction limit *under-states* the radius. The limit is optimistic rather than conservative, and it can call a market stable that is unstable at any finite correction strength. On uniform random draws with $\theta \in [0.01, 1]$ the limit calls 11.8% of configurations stable that are not. Three worked cases, all of which the limit reads as comfortably stable:

	Configuration	limit reads	truly unstable unless
708	$N = 10, N_b = 6, m_1 = 0.15, \kappa = 0.8, s = 1$	0.750	$\gamma_{PO} > 1.9\gamma$
	$N = 30, N_b = 8, m_1 = 0.10, \kappa = 0.9, s = 0.8$	0.604	$\gamma_{PO} > 3.9\gamma$
	$N = 20, N_b = 10, m_1 = 0.12, \kappa = 0.8, s = 1$	0.984	$\gamma_{PO} > 58.6\gamma$

Certificate C18 finds zero violations of Proposition E.1 on 3000 random configurations, each swept over 50 values of θ , and reports the verdict flips above. This is why the paper states the exact two-block root as the theorem and the limit as a corollary carrying its own error direction: shipping the limit alone would state a stability criterion that errs in the unsafe direction without saying so.

E.2 Imperfect correction

Proof of the imperfect-correction law and of Corollary 5. At $\kappa = s = 1$ the quadratic degenerates, since $1 - k_s = 0$ kills a, b and Q , leaving $\rho = P = m_1(N_b + \theta N_{\text{corr}})$. Writing the corrected fraction as $\rho_c = 1 - N_b/N$ and solving $\rho < 1$ for it gives

$$\rho^*(\theta) = \frac{1 - 1/m_N}{1 - \theta} = \frac{1 - 1/m_N}{e}, \quad e = 1 - \theta = 1 - \gamma/\gamma_{PO},$$

which is the standard epidemiological coverage requirement for an imperfect vaccine, with the clean law $1 - 1/m_N$ as its perfect-efficacy corner. The required fraction exceeds one, so that no corrected fraction stabilizes the market, exactly when $\theta > 1/m_N$, that is when $\gamma_{PO}/\gamma \leq m_N$, which is Corollary 5. $\square$

Certificate H5 reports agreement to 1.8×10^{-15} over 120 configurations, with the radius form checked against dense eigensolves, and H6 certifies the critical efficacy. At $m_N = 2.5$ the corrected update

must deliver at least a $2.5\times$ curvature improvement or the market cannot be stabilized by un-blinding at all, whatever fraction is treated. This is the exact structural parallel of the critical crowding level of Appendix D: each lever has a regime past which it stops working, and stating both is what makes the substitution frontier an honest object rather than an extrapolation.

The correspondence is worth being precise about. It now transfers not only the threshold but the standard refinement of the threshold, which is what a structural analogy predicts and a decorative one does not. Away from $\kappa = s = 1$ the closed form is no longer exact and the exact threshold comes from setting the quadratic's larger root to one; the law remains accurate to well under one firm's granularity at the parameters this paper uses.

E.3 The threshold in firms

Proposition E.2. The strong-correction limit is stable if and only if $N_b < N_c(s) = 1 + (1/m_1 - 1)/(\kappa s)$.

Proof. Rearrange $m_1(1 + \kappa s(N_b - 1)) < 1$. □

Certificate C14 finds zero mismatches on 4000 draws for this form. The clamped restatement is *not* equivalent: writing $\rho_c = 1 - N_b/N$ and $\rho^* = \max(0, 1 - N_c(s)/N)$, the criterion $\rho_c > \rho^*$ fails on 134 of those same 4000 draws, every failure at the same corner, namely $N_b = N$ with $N_c(s) \geq N$, the all-blind market that is stable because it needs no correction at all and that a clamp at zero combined with a strict inequality excludes. The paper therefore states $N_b < N_c(s)$ as the theorem, which is primitive and exact, and presents the clamped ρ^* as the policy object, read as the fraction that must be corrected, with $\rho^* = 0$ meaning no requirement rather than a strict inequality to clear.

Since ρ^*N is generally not an integer and the panels run at N in the tens, where one firm is several percentage points, the realized requirement is

$$\text{minimum corrected firms} = N - \lceil N_c(s) \rceil + 1,$$

clamped to $[0, N]$, because the largest stable blind count is $\lceil N_c \rceil - 1$. The natural guess $\lceil \rho^*N \rceil$ agrees on every random draw, since random draws never land on an exact integer N_c , but it is off by one exactly when N_c is an integer: at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$ and the s giving $N_c = 8$, the truth is 13 corrected firms and the guess says 12. Certificate H4 checks the correct form exactly on 3000 draws. Worked at $N = 20$, $m_1 = 0.15$ and $\kappa = 0.8$:

s	$N_c(s)$	ρ^*	minimum corrected firms
1.0	8.08	0.596	12
0.5	15.17	0.242	5
0.2	36.42	0	0

At $\kappa = s = 1$ we have $N_c = 1/m_1$ and $N_{\text{eff}} = N$, so $\rho^* = 1 - 1/(m_1N) = 1 - 1/m_N$, the epidemiological threshold with the systemic modulus as the reproduction number: a market of ten firms at $m_N = 2.5$ needs 60% un-blinded. And ρ^* is increasing in s , strictly so wherever it is positive, which is what makes model diversity and corrected learning substitutes and what the (ρ, s) frontier plots. Certified in C15 and C16.

Panel 4 runs at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$ and $s = 1$, where the imperfect-correction law is not exact, and measures thresholds of 12, 14, 16 and 20 firms at efficacies 1.00, 0.90, 0.75 and 0.60 against a law predicting 12, 14, 16 and 20. That is four out of four at the integer granularity a market actually faces. It is a dry run in the linearized reference environment and is recorded as evidence rather than as proof; the exact statement away from $\kappa = s = 1$ remains the quadratic's root.

F The Pigouvian wedge

F.1 The welfare object

Drive the linearized joint map with noise. Under (A1) and (A5) the binding mode is the common one, whose deviation z_t from the joint equilibrium obeys $z_{t+1} = -m_N z_t + \xi_t$ with ξ_t independent,

mean zero and variance σ^2 . This is an AR(1) with coefficient $-m_N$. Stationarity requires $|m_N| < 1$, which is the stability condition the rest of the paper works with, and taking variances of a stationary solution gives $V = m_N^2 V + \sigma^2$, hence

$$V(m_N) = \frac{\sigma^2}{1 - m_N^2}.$$

The coefficient is $-m_N$ and not m_N because the retraining Jacobian is $J = -m_1[(1 - \kappa)I + \kappa R]$ and the common mode inherits its sign. The stationary variance is blind to that sign, since the recursion enters squared, which is why the same expression serves both. The sign is not lost information: it reappears in the lag-1 autocorrelation, which is $-m_N$ and therefore negative, so a crowded market shows oscillatory rather than persistent common-mode error. That is the observable the supervision estimator reads, and it is why a positive-autocorrelation diagnostic would look for the wrong thing. Certificate C19 checks both facts by simulation rather than asserting them.

V is the paper's price of instability, and it is a modeling choice: stationary variance is the smooth proxy for divergence risk, finite everywhere strictly inside the stable region and divergent exactly at the boundary the rest of the paper is about. Nothing below depends on the functional form beyond V being positive, increasing and convex in m_N on $[0, 1)$, all three of which the closed form satisfies.

(W1) Aggressiveness. Each firm chooses $a_i > 0$, meaning cadence, learning rate or responsiveness: anything that scales its own modulus. Write $m_i = \mu(a_i)$ with μ differentiable, strictly increasing and common across firms. Aggressiveness moves a firm's gain and not its response direction, so R and κ are held fixed.

(W2) Benefit. Aggressiveness buys tracking benefit $B(a_i)$, strictly increasing and strictly concave. Only firm i 's own benefit depends on a_i , so the entire interaction runs through m_N .

(W3) Exchangeable alignment. R is invariant under a common permutation of firms. The supply-chain alignment is exchangeable, as are the monoculture and orthogonal corners. This buys the uniform leading eigenvector at the symmetric point.

(W4) Client exposure. The planner's variance weight satisfies $W > \sum_i w_i$, with all weights positive.

(W4) is stated as an assumption and is not microfounded. Every trade has a client on the other side who bears execution-quality variance while no dealer's objective contains it, so the planner's weight on common-mode variance strictly exceeds what the dealers together internalize. The results below turn on the sign of $W - \sum_i w_i$ and never on its size, so a microfoundation would fix a magnitude the theorem does not use.

F.2 The marginal crowding share

Proof of Lemma 6. By Proposition B.6, for $M = \text{diag}(m_1, \dots, m_N)$ and $B = (1 - \kappa)I + \kappa R$ the radius of $J = -MB$ is $\lambda_{\max}(A)$ for $A(M) = M^{1/2}BM^{1/2}$. Since $dM^{1/2}/dm_i = (1/(2\sqrt{m_i}))e_i e_i^\top$,

$$\frac{dA}{dm_i} = \frac{1}{2\sqrt{m_i}} \left(e_i e_i^\top B M^{1/2} + M^{1/2} B e_i e_i^\top \right).$$

For a simple eigenvalue with unit eigenvector v , first-order perturbation gives $d\lambda_{\max}/dm_i = v^\top (dA/dm_i) v$. The two terms are transposes of one another and A is symmetric, so they contribute equally, giving $d\lambda_{\max}/dm_i = (BM^{1/2}v)_i v_i / \sqrt{m_i}$. At the symmetric point $M = m_1 I$ we have $A = m_1 B$, so v is the leading eigenvector of B and $BM^{1/2}v = \sqrt{m_1} \lambda_{\max}(B)v = \sqrt{m_1} N_{\text{eff}} v$, whence $d\lambda_{\max}/dm_i = N_{\text{eff}} v_i^2$. Summing over i gives N_{eff} , because v is a unit vector. Under (W3) an exchangeable B commutes with every permutation matrix, so a simple leading eigenvector must be permutation invariant, giving $v_i^2 = 1/N$ and $dm_N/dm_i = N_{\text{eff}}/N$. $\square$

Certificate C34 checks this by finite differences on random alignment matrices, confirming that the shares sum to N_{eff} and equal N_{eff}/N on exchangeable R . It is the load-bearing certificate of this section: everything else in the theorem is arithmetic on top of the marginal crowding share, and until C34 that share was the step where a plausible-looking $1/N$ could have been wrong by a factor of N_{eff} .

Two things are worth reading off. The *sum* of the marginal effects is N_{eff} and not 1, so the market's total sensitivity to a uniform increase in aggressiveness is amplified by exactly the effective learner count. And an *individual* firm moves m_N by N_{eff}/N , so as N grows each firm's own footprint shrinks while the total grows, which is the arithmetic shape of every commons problem.

F.3 The two problems and the wedge

Firm i bears weight w_i on common-mode variance and the planner bears W :

$$\text{private: } \max_{a_i} B(a_i) - w_i V(m_N(a)), \quad \text{social: } \max_a \sum_j B(a_j) - W V(m_N(a)).$$

Write $V'(m) = 2\sigma^2 m / (1 - m^2)^2 > 0$ on $(0, 1)$ and evaluate at a symmetric profile $a_i = a$, so that $m_N = \mu(a)N_{\text{eff}}$. Chaining (W1) with Lemma 6 gives $dm_N/da_i = (N_{\text{eff}}/N)\mu'(a)$ at the symmetric profile, and the two first-order conditions are

$$(P) \quad B'(a) = w_i V'(m_N) \frac{N_{\text{eff}}}{N} \mu'(a), \quad (S) \quad B'(a) = W V'(m_N) \frac{N_{\text{eff}}}{N} \mu'(a).$$

They are the same equation with a different price on crowding, and the theorem is the gap between the two coefficients.

Proof of Theorem 7. Subtract (P) from (S). A firm facing a per-unit fee t on aggressiveness solves $\max B(a_i) - w_i V(m_N(a)) - ta_i$, whose first-order condition is (P) with $B'(a)$ replaced by $B'(a) - t$. Substituting

$$t^* = (W - w_i) V'(m_N) \frac{N_{\text{eff}}}{N} \mu'(a) = (W - w_i) \frac{2\sigma^2 m_N}{(1 - m_N^2)^2} \cdot \frac{N_{\text{eff}}}{N} \mu'(a)$$

reproduces (S) term for term, so the taxed decentralized equilibrium implements the symmetric social optimum. $\square$

With symmetric weights $w_i = w$ and $W = \chi N w$ for the $\chi > 1$ that (W4) supplies, $t^* = (\chi N - 1) w V'(m_N) (N_{\text{eff}}/N) \mu'(a)$ against a total marginal cost of $\chi N w V'(m_N) (N_{\text{eff}}/N) \mu'(a)$, so the ignored fraction is $(\chi N - 1)/(\chi N)$. Setting $\chi = 1$, which shuts off client exposure entirely and leaves only the firm-to-firm channel, the ignored fraction is exactly $(N - 1)/N$. Note that N_{eff} cancels from this ratio, so the $(N - 1)/N$ reading is a generic commons result and it is the provenance asymmetry below that is specific to this paper.

Proposition F.1. t^* is strictly increasing in N , in κ and in m_N , and diverges as $m_N \rightarrow 1$.

Proof. Take the supply-chain specialization $N_{\text{eff}} = 1 + \kappa s(N - 1)$ with symmetric weights, so that $t^* = (\chi - 1/N) w V'(m_N) N_{\text{eff}} \mu'(a)$. The factor $\chi - 1/N$ is strictly increasing in N ; N_{eff} is strictly increasing in N whenever $\kappa s > 0$ and in κ whenever $s > 0$ and $N \geq 2$; $m_N = \mu(a)N_{\text{eff}}$ inherits both; and $V'(m) = 2\sigma^2 m / (1 - m^2)^2$ is strictly increasing on $(0, 1)$ and diverges at 1, since both m and $(1 - m^2)^{-2}$ are. Every factor moves the same way, so the product does. $\square$

The divergence is the economically loaded part. The fee that would correct a market is not proportional to how crowded it is; it blows up like $(1 - m_N)^{-2}$ as the market approaches its stability boundary, so a regulator setting a fee from a comfortable-looking market and holding it fixed is setting it far too low by the time the market is close to the edge. Certified in C20.

Proof of Corollary 8. Write $C(a) = V(m_N(a))$ and

$$F_{\text{priv}}(a) = B'(a) - w C'(a), \quad F_{\text{soc}}(a) = B'(a) - W C'(a).$$

Both are strictly decreasing, since B' is strictly decreasing by (W2) and C' is nondecreasing by convexity, so each has at most one zero and by the interior assumption exactly one, namely a_d and a_s . Because $C'(a) > 0$ on the stable range, as $V' > 0$, $N_{\text{eff}} > 0$ and $\mu' > 0$, and because $W > w$ by (W4),

$$F_{\text{soc}}(a) = F_{\text{priv}}(a) - (W - w) C'(a) < F_{\text{priv}}(a) \quad \text{for every } a.$$

848 Evaluating at a_d gives $F_{\text{soc}}(a_d) < F_{\text{priv}}(a_d) = 0$, and since F_{soc} is strictly decreasing and vanishes
849 at a_s , this forces $a_s < a_d$. Setting $\chi = 1$ so that $W = Nw$ gives $W - w = (N - 1)w > 0$ exactly
850 when $N \geq 2$, so the strict inequality survives with no client-side channel at all. At $N = 1$ with $\chi = 1$
851 the wedge is zero and the two problems coincide, which is the correct degenerate case: a single firm
852 bearing its own variance in full internalizes everything. $\square$

853 The distortion grows as the market crowds, since the coefficient gap $(W - w)C'(a)$ is proportional
854 to $V'(m_N)N_{\text{eff}}$, both of which increase in N and in κ . Certified in C21, including the vanishing gap
855 at $N = 1$.

856 F.4 The provenance channel

857 **Proposition F.2.** Under the supply-chain specialization, $dm_N/ds = m_1\kappa(N - 1)$ exactly, and the
858 corresponding adoption wedge is $t_s^* = (W - w_i)V'(m_N)m_1\kappa(N - 1)$.

859 *Proof.* $m_N = m_1(1 + \kappa s(N - 1))$ is affine in s , so the derivative is the coefficient, and the wedge
860 follows from the proof of Theorem 7 with s in place of a_i as the choice variable. $\square$

861 The same wedge therefore prices shared-model adoption directly, which turns the monoculture
862 externality into a fee rather than a warning. The contrast with the aggressiveness channel is the
863 paper-specific result. The aggressiveness derivative carries a factor N_{eff}/N , bounded above by
864 $(1 + \kappa(N - 1))/N$ and therefore tending to κ from above as N grows, so a firm’s marginal footprint
865 through its own cadence does not grow without bound. The provenance channel is linear in N and
866 does not decay. A firm’s decision to adopt the market-leading vendor imposes a marginal cost on
867 the market that grows with market size while its private quality gain does not, and that asymmetry is
868 why unregulated adoption dynamics run toward the unstable configuration rather than away from it,
869 and why a diversity floor is a different instrument from a cadence cap rather than a restatement of it.
870 Certificate W6 measures the provenance channel growing from 0.79 to 87.91 between $N = 10$ and
871 $N = 1000$ while the aggressiveness channel falls from 0.82 toward $\kappa = 0.8$.

872 The three levers are the standard instrument taxonomy: cadence caps are quantity regulation, correc-
873 tion mandates are a technology mandate, diversity floors are a structural remedy, and the wedge is a
874 price instrument. The choice between prices and quantities under uncertainty is Weitzman’s question,
875 which this paper cites for framing rather than solves.

876 G Supervision from public prices

877 Every quantity in Theorems 2 through 7 involves R and s , which no outsider observes. Supervisors
878 do not inspect firms’ models, training sets or vendor contracts, so a theory whose every variable is
879 unobservable to the regulator it addresses is a theory of a problem rather than a theory a regulator can
880 act on.

881 Near the boundary the system tells on itself. As $m_N \rightarrow 1$ the aligned combination of firms’ decisions
882 exhibits critical slowing down: the lag-1 autocorrelation of the leading principal component of public
883 quotes approaches one in magnitude, negative because the common mode oscillates, and its variance
884 share grows like $1/(1 - m_N)$. A supervisor can therefore estimate distance to instability from public
885 prices alone, with no access to any firm’s model, data or code, and the estimator sharpens exactly
886 as the market approaches the boundary, which is the opposite of the usual situation in which risk
887 measurement degrades in the regime that matters.

888 This material is deliberately thin and the body flags it as deferred. The body carries the estimator
889 and the consistency claim in one paragraph. The empirical panel, namely the leading component’s
890 variance share and lag-1 autocorrelation of cross-sectional spread co-movement overlaid on the
891 model-implied fragility index, together with a placebo on series where no dealer-model channel
892 exists, is scoped and is not run for this submission. It would be framed as consistency evidence
893 and never as identification, since spread co-movement is contaminated by common macro shocks
894 and the data are public proxies rather than trade-level records. The consistency proof, the central
895 limit theorem and the detection sample complexity are journal material. Writing the estimator down
896 formally, rather than as the description above, is the first open item.

H Certificates

Every closed form in the body carries a numerical certificate. A certificate is an assertion-based script that re-derives the expectation independently, by dense eigensolve, finite difference, brute force or direct simulation, rather than importing the theory module’s own answer, and that exits nonzero on any disagreement. A closed form reaching the paper without a passing certificate ships as derived and never as verified.

Assertions	File	What it establishes
123	<code>verify_theorem1_proof.py</code>	the reduction, the properties of R , the spectrum, the mean bound, the heterogeneous-modulus bound, and the supply-chain concentration rate
60	<code>verify_theorem1_anchors.py</code>	the three anchors, the clustered topology, and the signed error of the mean index
59	<code>verify_theorem2_cadence.py</code>	the inner contraction, hypothesis (C), the joint map, the frontier in both forms, and the realized dynamics
70	<code>verify_theorem3_herd_immunity.py</code>	the two-block quadratic, the degenerate blocks, monotonicity in θ , the imperfect-correction law, the critical efficacy and the integer threshold
125	<code>verify_theorem4_wedge.py</code>	the AR(1) reduction and its sign convention, Lemma 6 by finite differences, the comparative statics, and over-adaptation
56	<code>verify_theory_module.py</code>	acceptance tests on the closed-form module
32	<code>verify_hetero_env.py</code>	the reference environment, including its reduction to the homogeneous market at $R = \mathbf{11}^\top$
525	total	all passing at submission

Two of these were falsifying rather than confirming. C18 showed that the strong-correction limit errs optimistic, which changed what Theorem 4 claims and moved the exact two-block root from an optional refinement to the theorem itself. C14 showed that the clamped restatement of the blind-firm threshold is not equivalent to the primitive form, failing on 134 of 4000 draws at the all-blind corner. Both corrections are recorded above, where they apply.

I Experimental specifications

Environments. There are two, and the distinction between them is what the status flags track. The *order-flow simulator* is a genuine N -dealer market with informed flow, spread and inventory, which reduces bit for bit to the single-dealer market at $N = 1$. The *reference environment* places firms at chosen response directions and lets them retrain against a shared pool, realizing the response geometry of Lemma 1 and nothing else: no informed flow, no spread, no inventory, no microstructure of any kind. Its acceptance test, run before any measurement is taken from it, is that at $R = \mathbf{11}^\top$ it reproduces the homogeneous market on three counts, namely a measured alignment of exactly $\mathbf{11}^\top$, an N_{eff} of exactly $1 + \kappa(N - 1)$, and a built Jacobian equal to (1) to 10^{-10} .

Protocol. Inherited from the base model and non-negotiable, since violating any of these silently invalidates a panel: sweep the feedback gain and never the confounded adversariality parameter; probe at the operating spread with common random numbers; scale the liquidity boost per the environment’s guidance when multi-dealer runs approach the informed-flow cap, and never de-saturate silently; and treat beyond-boundary probe readings as diagnostics rather than as slopes. All runs are CPU-only and deterministic from a (config, seed) pair. Stability is estimated from the trajectory’s asymptotic growth rate under the actual retraining map rather than read off an eigensolve, so a panel can disagree with its closed form.

The panels. Panel 1 sweeps N in the homogeneous shared-pool market and reproduces the published amplification, which is external validation against a prior run rather than a self-consistency check, and it is the paper’s one measured result. Panel 2 measures m_N over a grid of firms by shared-model fraction against the predicted boundary, with firms placed at the target alignment exactly rather than in distribution so that an s sweep moves along the exact curve; its clustered companion, three aligned firms among ten, is the topology worked in Appendix B. Panel 3 sweeps the (N_{eff}, K) grid

932 against $K_{\max}$ at $c = 0.8$, including the critical-crowding column where the window closes. Panel
 933 4 runs mixed markets at corrected fractions $0, 1/N, \dots, 1$ in an unstable market. Panel 5 traces
 934 the (ρ, s) iso-stability curve, and it is the paper’s most policy-legible object. Panel 6 solves both
 935 first-order conditions by bisection with m_N read from the reference environment’s realized dynamics
 936 rather than from the closed form, so the over-adaptation verdict is a property of the simulated market.

937 **Why five of six are dry runs.** A heterogeneous-response port of the order-flow simulator was built
 938 and is exact at flat exposure profiles, reproducing the unmodified simulator at every N from 1 to 6
 939 at relative error 0.00. Its acceptance gate then failed. The liquidity and price-impact channels stay
 940 shared whatever the response profiles are, leaving a residual reaching 17.7 percentage points across
 941 the separation range at $N = 2$, larger than several effects these panels resolve, and at $N \geq 4$ the
 942 finite-difference probe no longer measures a local slope. Two repairs are identified, routing liquidity
 943 and impact through the profiles as well, or finding a configuration whose probe stays local at the
 944 N the panels need, and neither is attempted for this submission, since both are multi-day edits and
 945 the first risks the bit-for-bit anchor that panel 1 rests on. No figure is drawn from that port and no
 946 status was moved. Panel 6 is a dry run for a different reason: the order-flow simulator carries no
 947 aggressiveness choice variable and no welfare object, so a measured counterpart does not exist to be
 948 built.

949 J Deferred items

950 Stated so that the boundary of what is proved is legible. The fully heterogeneous case without (H1),
 951 which is the Kronecker object of Appendix B, and share-weighted alignment for unequal-sized firms,
 952 which needs a majorization condition on the share vector since the naive claim that any reweighting
 953 raises $\lambda_{\max}$ is false. Non-separable spillover κ_{ij} , which folds into R only when it factors. The sharp
 954 operator-norm rate $\sqrt{N}/d$ of Appendix C, which needs a matrix-Bernstein argument rather than the
 955 Frobenius step, and a version of design assumption (D) under dependence across the idiosyncratic
 956 components, which is what shared data rather than shared models would produce. Asynchronous
 957 and heterogeneous cadences, which break the eigenvector sharing that Theorem 3 uses. Three or
 958 more correction levels, which break the two-block collapse and return a genuine secular equation. A
 959 corrected firm that also changes its response direction, which would move R and which hypothesis
 960 (M) does not model. And beyond the symmetric point, the observation that $dm_N/dm_i = N_{\text{eff}} v_i^2$
 961 makes the marginal crowding share the leading eigenvector’s participation ratio at firm i , so a firm
 962 sitting on the crowded mode imposes more than its $1/N$ share, and the asymmetric wedge that follows
 963 is a genuine open question.

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